

Change to polar coordinates and evaluate.

1. $\int_0^1 \int_0^x (x+y) dy dx$

$x: y:$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$dx dy = r dr d\theta$$

$$\int_{\theta=0}^{\pi/4} \int_{r=0}^{\sec \theta} (r \cos \theta + r \sin \theta) r dr d\theta$$

$$= \int_0^{\pi/4} \int_0^{\sec \theta} (\cos \theta + \sin \theta) r^2 dr d\theta$$

$$= \int_0^{\pi/4} (\cos \theta + \sin \theta) \left(\frac{r^3}{3} \right)_0^{\sec \theta} d\theta$$

$$= \frac{1}{3} \int_0^{\pi/4} (\cos \theta + \sin \theta) \sec^3 \theta d\theta$$

$$= \frac{1}{3} \int_0^{\pi/4} \left(\underline{\sec^2 \theta} + \frac{\sin \theta}{\cos^3 \theta} \right) d\theta$$

$$= \frac{1}{3} \left\{ \left[\tan \theta \right]_0^{\pi/4} + \int_0^{\pi/4} \frac{\sin \theta}{\cos^3 \theta} d\theta \right\}$$

$$= \frac{1}{3} \left\{ (1-0) + \int_1^{1/\sqrt{2}} \frac{1}{t^3} (-dt) \right\}$$

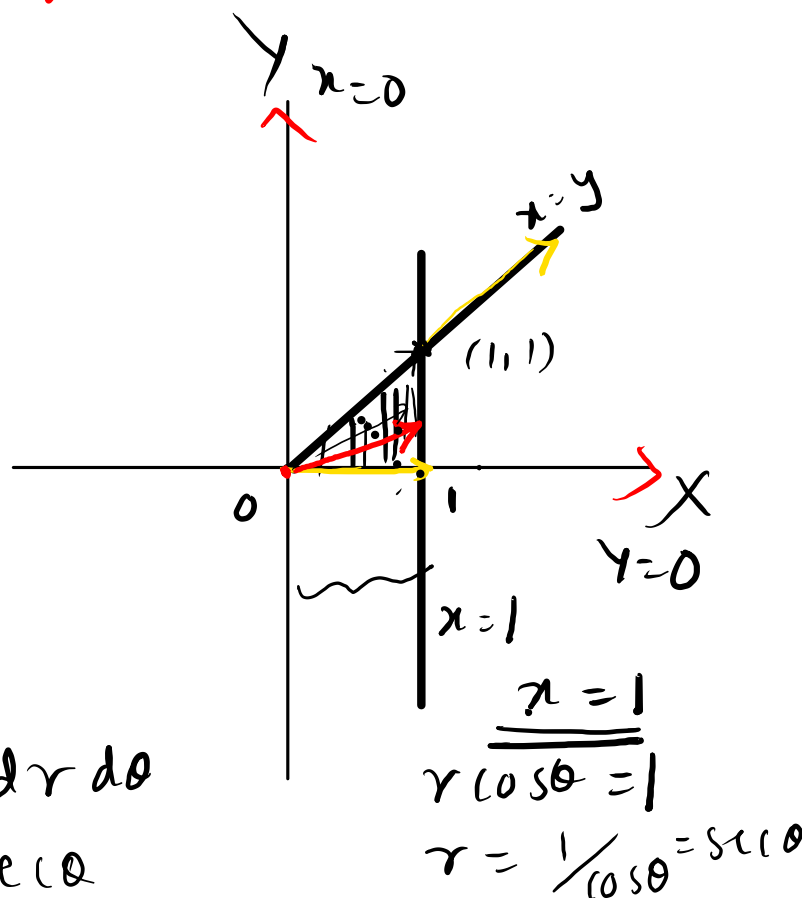
$$= \frac{1}{2}$$

$$x=0$$

$$y=0$$

$$x=1$$

$$y=x$$



$$\underline{x=1}$$

$$r \cos \theta = 1$$

$$r = \frac{1}{\cos \theta} = \sec \theta$$

$$\cancel{\cos \theta} \frac{1}{\cos^3 \theta}$$

$$\cos \theta = t$$

$$-\sin \theta d\theta = dt$$

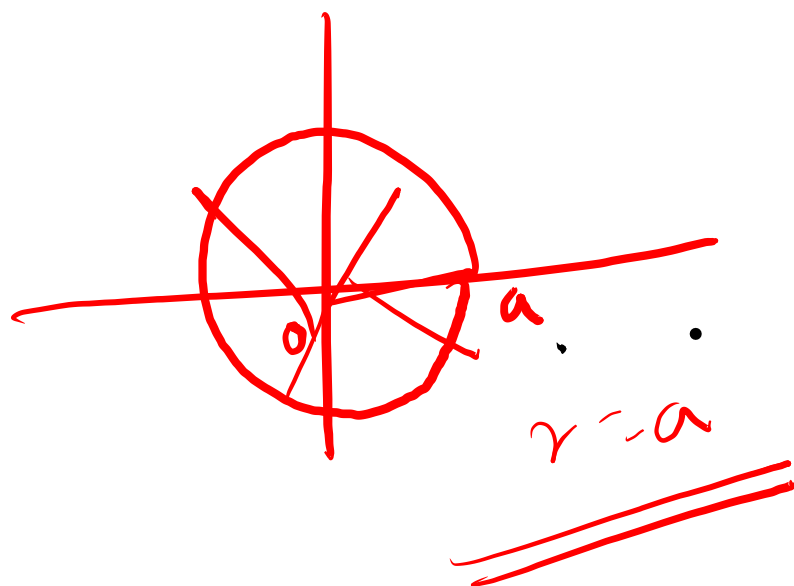
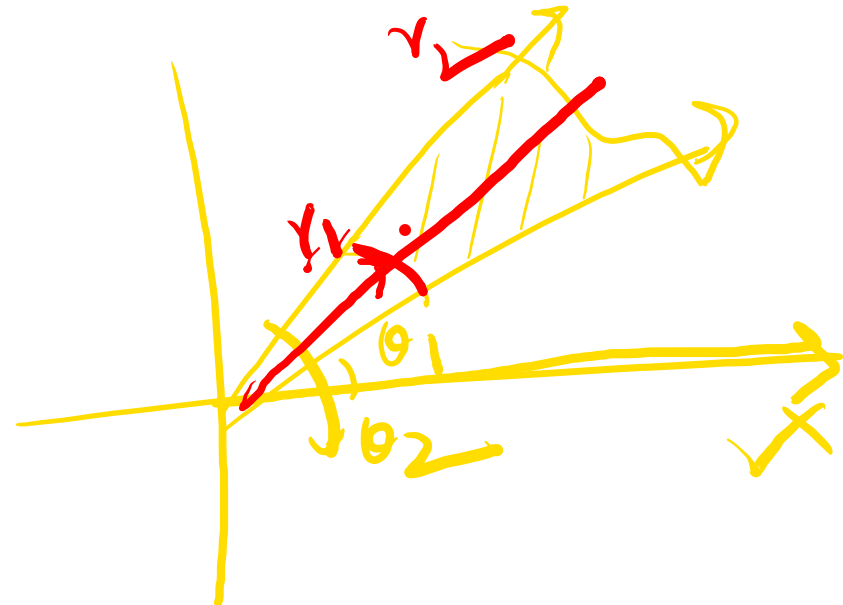
$$\sin \theta d\theta = -dt$$

$$\theta: 0 \rightarrow \pi/4$$

$$t: 1 \rightarrow 1/\sqrt{2}$$

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

$$\underline{\underline{dx dy = r dr d\theta}}$$

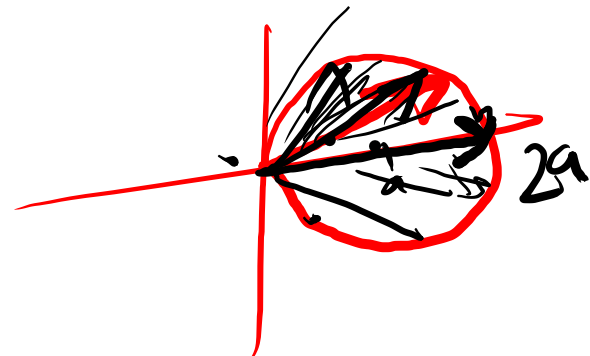


$$\theta: \theta_1 \longrightarrow \theta_2$$

$$r: r_1 \longrightarrow r_2$$

$$r_1 = f_1(\theta)$$

$$r_2 = f_2(\theta)$$



$$2. \int_0^1 \int_0^{\sqrt{1-y^2}} (x^2 + y^2) dx dy$$

$y=$
 $x=$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$dx dy = r dr d\theta$$

$$\int_{\theta=0}^{\pi/2} \int_{r=0}^1 r^2 \cdot r dr d\theta = \pi/8$$

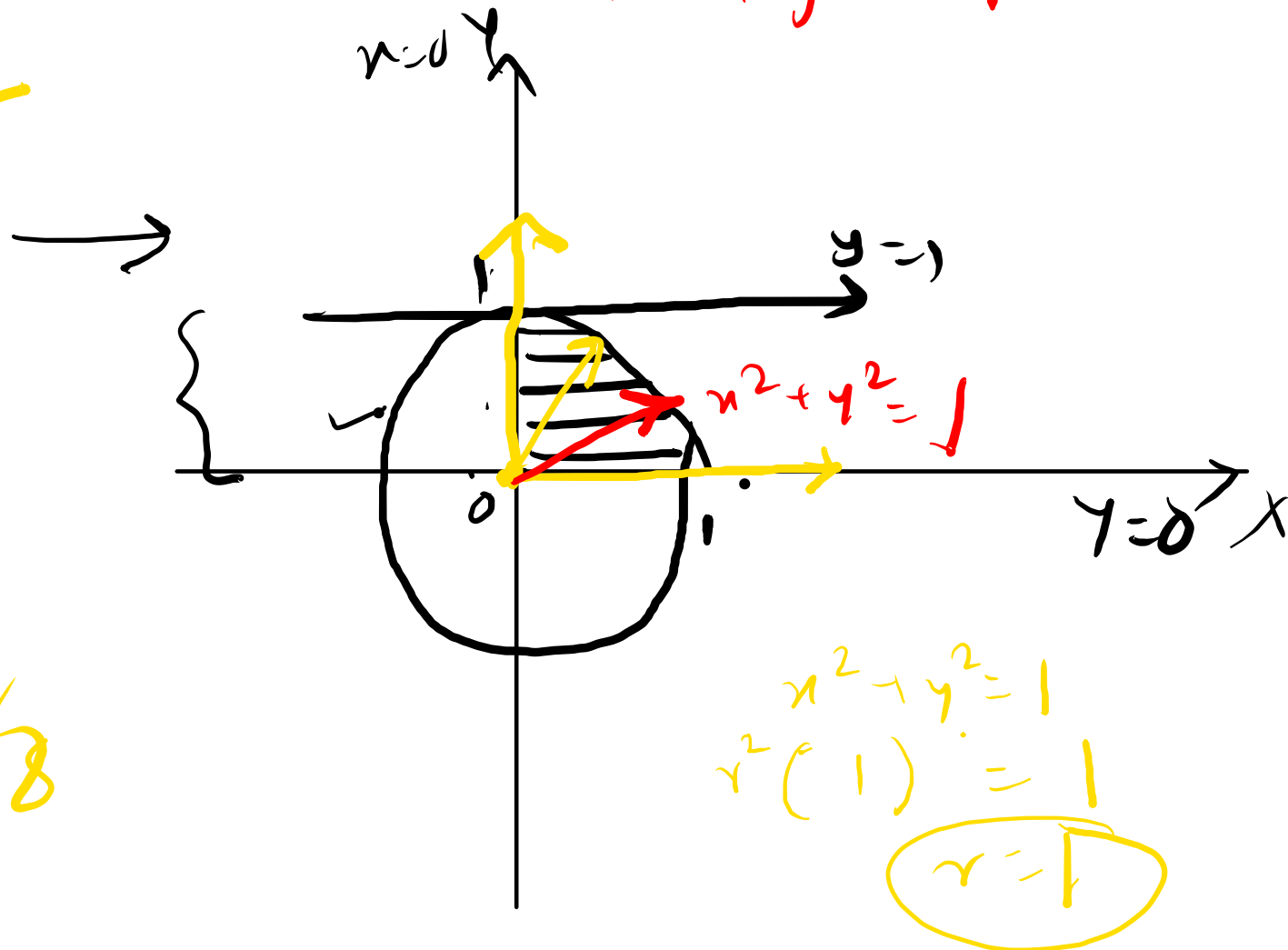
$$y=0$$

$$x=0$$

$$y=1$$

$$x=\sqrt{1-y^2}$$

$$x^2 + y^2 = 1$$



$$(x-a)^2 + (y-b)^2 = r^2$$

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$$4. \int_0^a \int_{\sqrt{ax-x^2}}^{\sqrt{a^2-x^2}} \frac{1}{\sqrt{a^2-x^2-y^2}} dy dx$$

$x=0$ $y=$

$x=0$

$x=a$

$$\begin{aligned} x &= r \cos \theta \\ y &= r \sin \theta \\ dx dy &= r dr d\theta \end{aligned}$$

$$\begin{aligned} y &= \sqrt{ax-x^2} \\ y^2 &= ax-x^2 \end{aligned}$$

$$\begin{aligned} y &= \sqrt{a^2-x^2} \\ y^2 &= a^2-x^2 \end{aligned}$$

$$\begin{aligned} x^2 + y^2 &= a^2 \\ (0,0) & \text{ rad } a \end{aligned}$$

$$\begin{aligned} x^2 - ax + \frac{a^2}{4} + y^2 &= \frac{a^2}{4} \\ (x - \frac{a}{2})^2 + y^2 &= (\frac{a}{2})^2 \end{aligned}$$

center $(\frac{a}{2}, 0)$, rad $\frac{a}{2}$

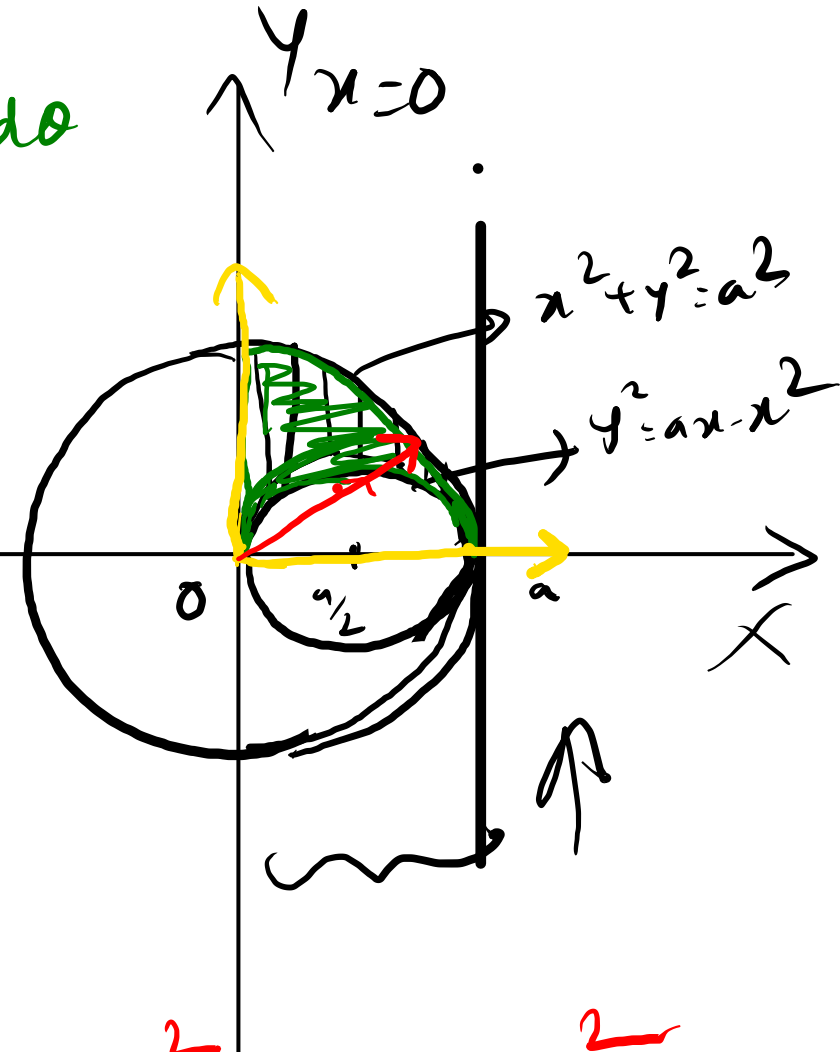
$$\int_0^{\pi/2} \int_{r=a \cos \theta}^a \frac{1}{\sqrt{a^2-r^2}} r dr d\theta$$

$= a$

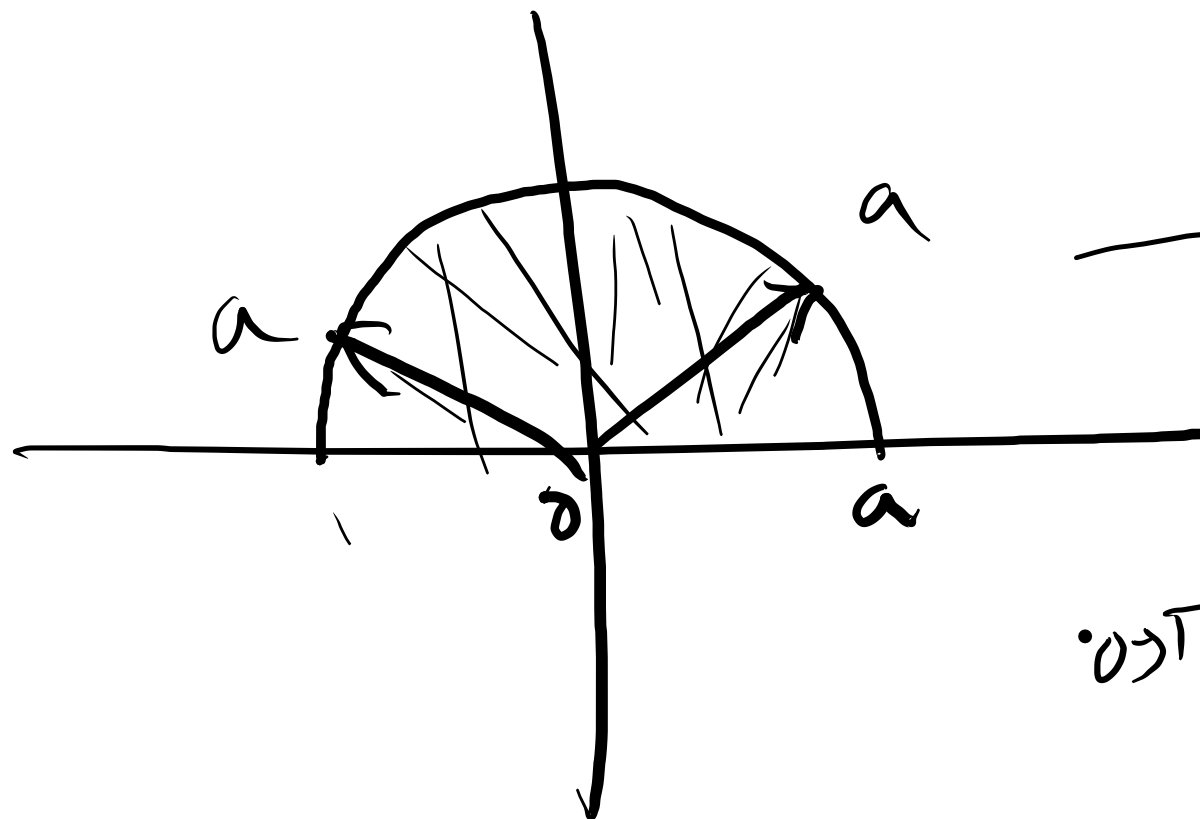
$$a^2 - r^2 = t$$

$$-2r dr = dt$$

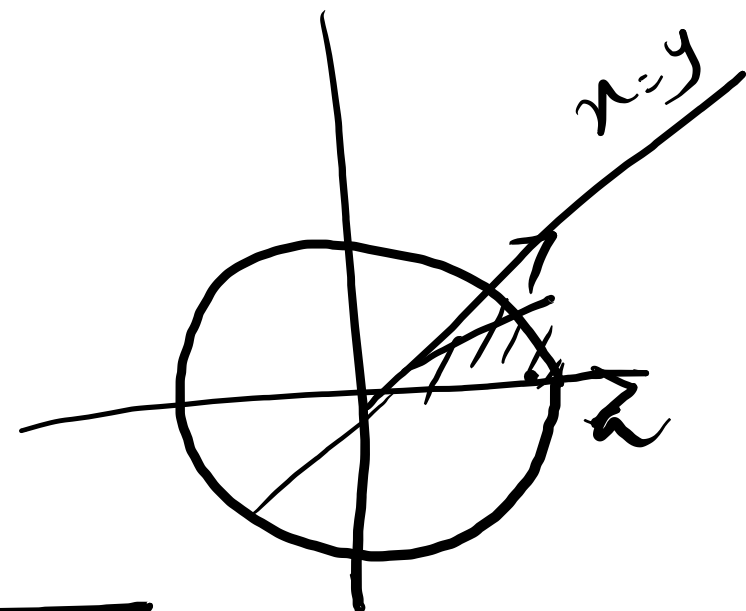
$$\frac{M^2}{4F}$$



$$\begin{aligned} y^2 &= ax - x^2 \\ x^2 + y^2 &= ax \\ r^2 &= ar \cos \theta \end{aligned}$$

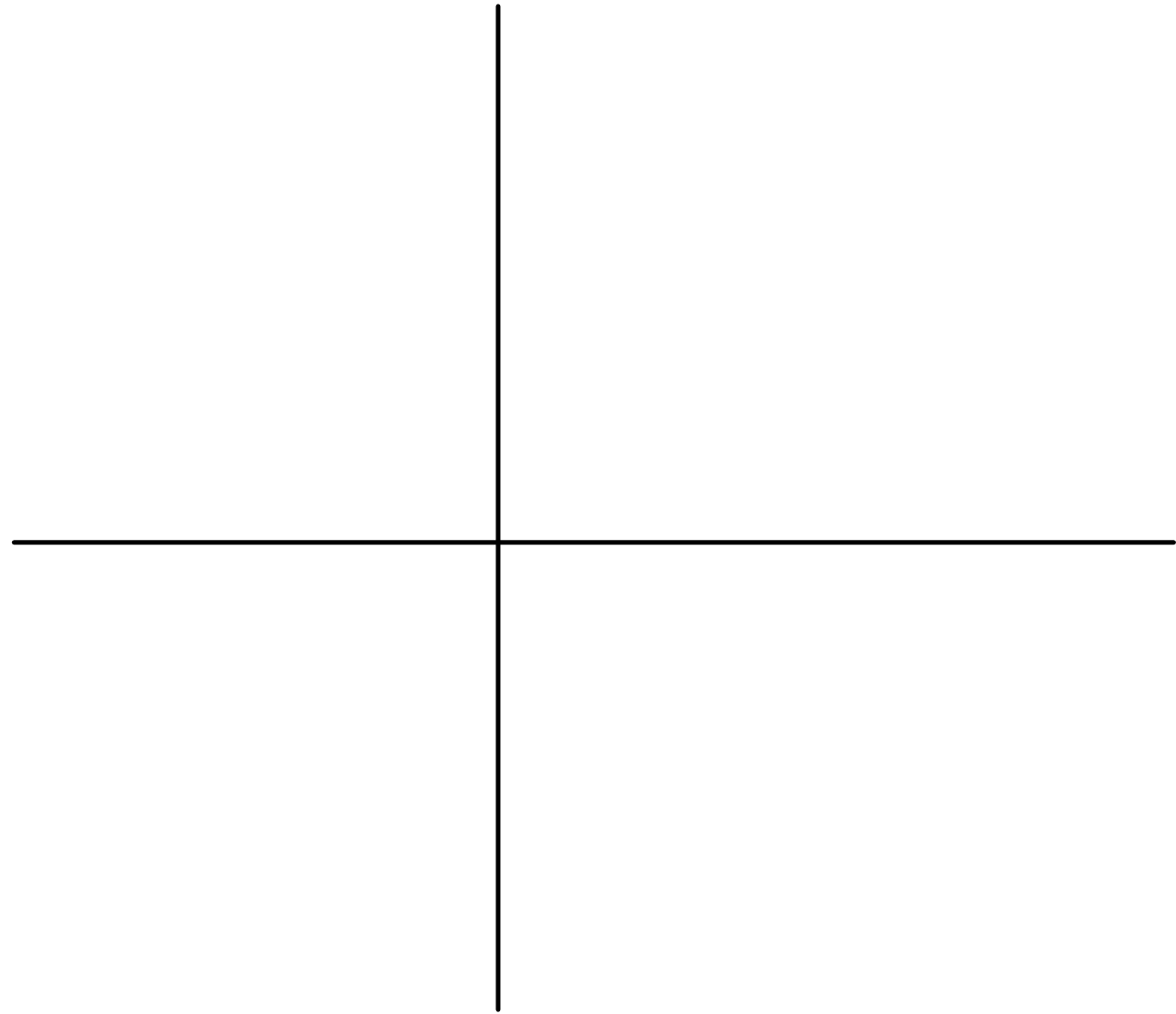


0.5π



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4.
$$\int_0^a \int_{\sqrt{ax-x^2}}^{\sqrt{a^2-x^2}} \frac{1}{\sqrt{a^2-x^2-y^2}} dy dx$$

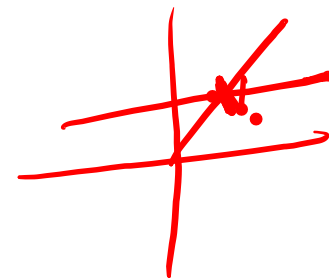


5. $\int_0^{a/\sqrt{2}} \int_y^{\sqrt{a^2-y^2}} \log(x^2 + y^2) dx dy$

$y = x$

$\sqrt{2}$

HW

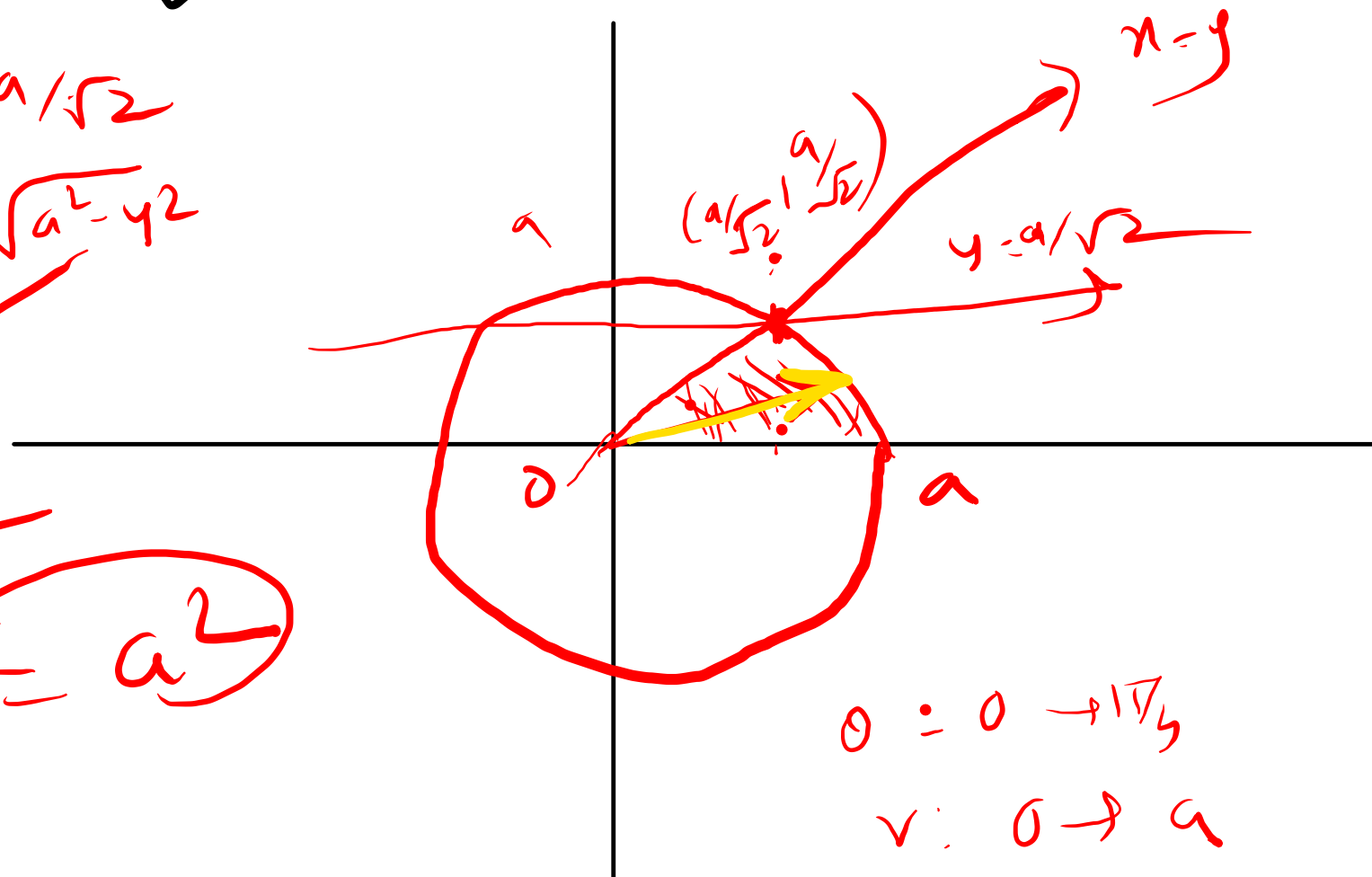


$y = 0$

$y = a/\sqrt{2}$

$x = \sqrt{a^2 - y^2}$

$x = y$



$x^2 + y^2 = a^2$
 $\frac{a^2}{2} + \frac{a^2}{2} = a^2$

$\theta: 0 \rightarrow \pi/4$
 $r: 0 \rightarrow a$

$2 \log r$

$$6. \int_0^{4a} \int_{y^2/4a}^y \left(\frac{x^2 - y^2}{x^2 + y^2} \right) dx dy$$

$y = x$

$$y = 0, \quad y = 4a$$

$$x = y^2/4a$$

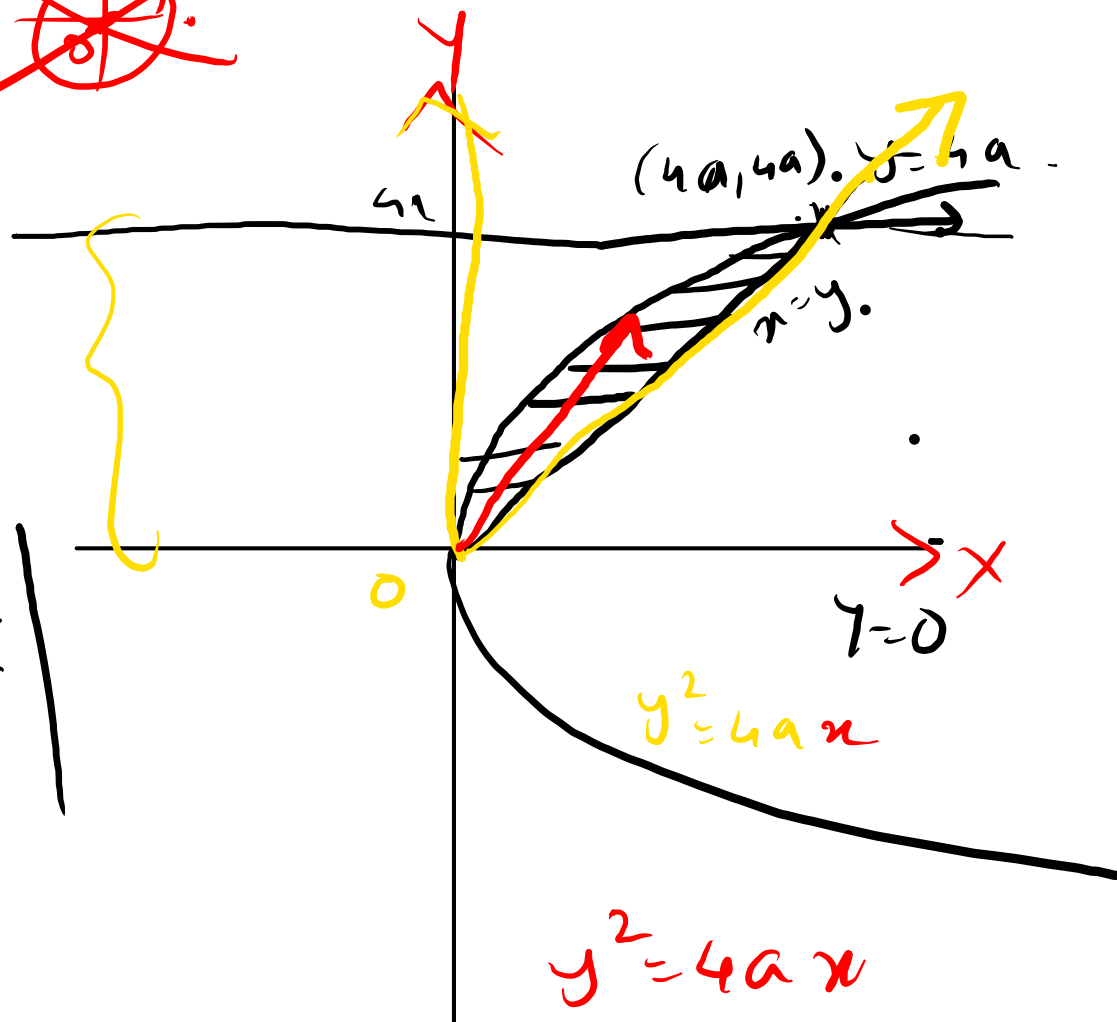
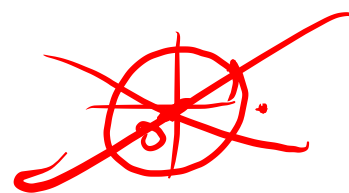
$$x = y$$

$$y^2 = 4ax$$

$$y = \pm \sqrt{4ax}$$

$$x = 4a$$

$$y = \pm \sqrt{16a^2} = \pm 4a$$



$$\theta: \frac{\pi}{4} \rightarrow \frac{\pi}{2}$$

$$r: 0 \rightarrow \frac{4a \cos \theta}{\sin^2 \theta}$$

$$\frac{\pi}{2} \quad \frac{4a \cos \theta}{\sin^2 \theta}$$

$$\int_{\pi/4}^{\pi/2} \int_0^{\frac{4a \cos \theta}{\sin^2 \theta}} \frac{r^2 (\cos^2 \theta - \sin^2 \theta)}{r^2} r dr d\theta$$

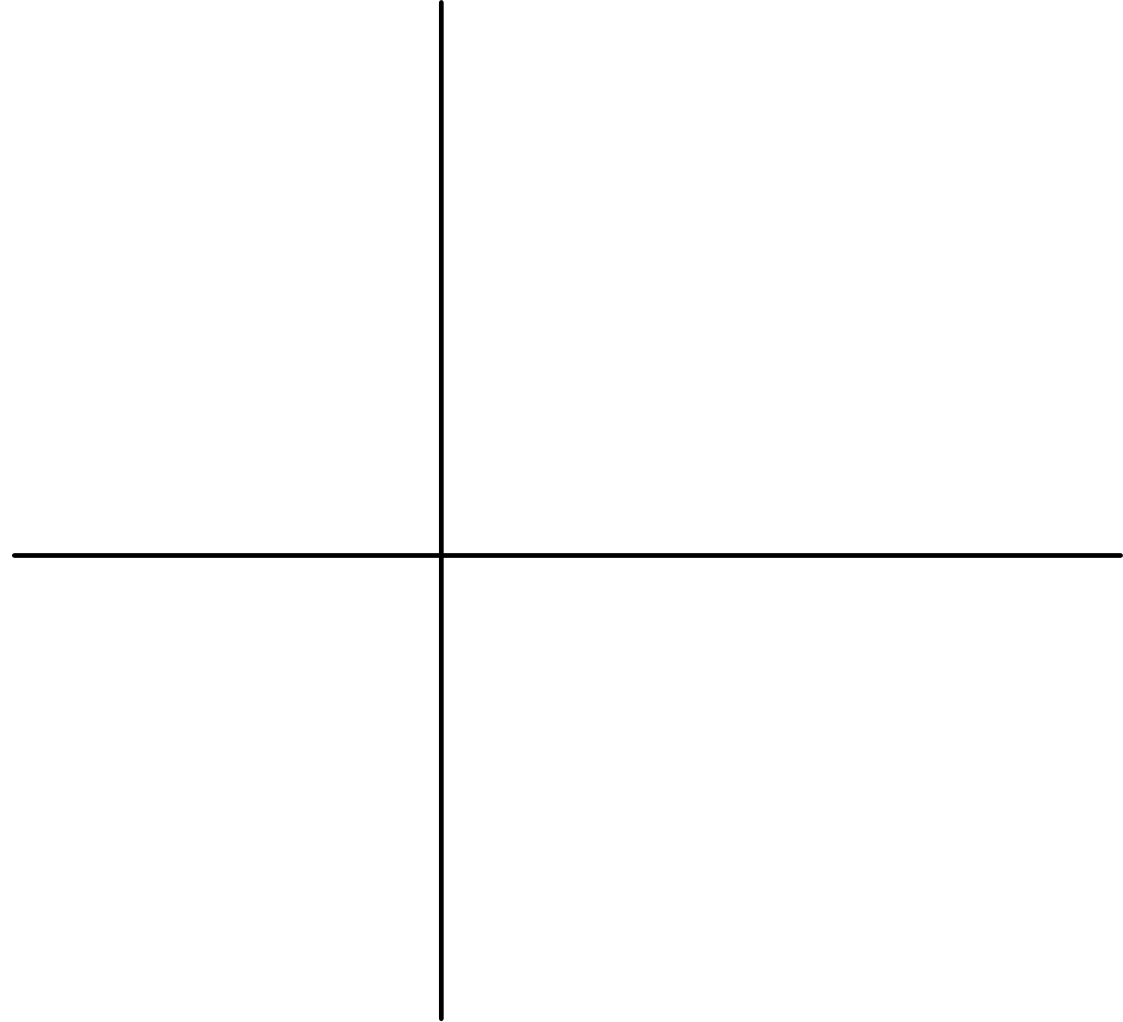
$$y^2 = 4ax$$

$$r^2 \sin^2 \theta = 4a r \cos \theta$$

$$r = \frac{4a \cos \theta}{\sin^2 \theta}$$

~~(4a, 4a)~~

7. $\int_0^a \int_{2\sqrt{ax}}^{\sqrt{5ax-x^2}} \frac{\sqrt{x^2+y^2}}{y^2} dy dx$

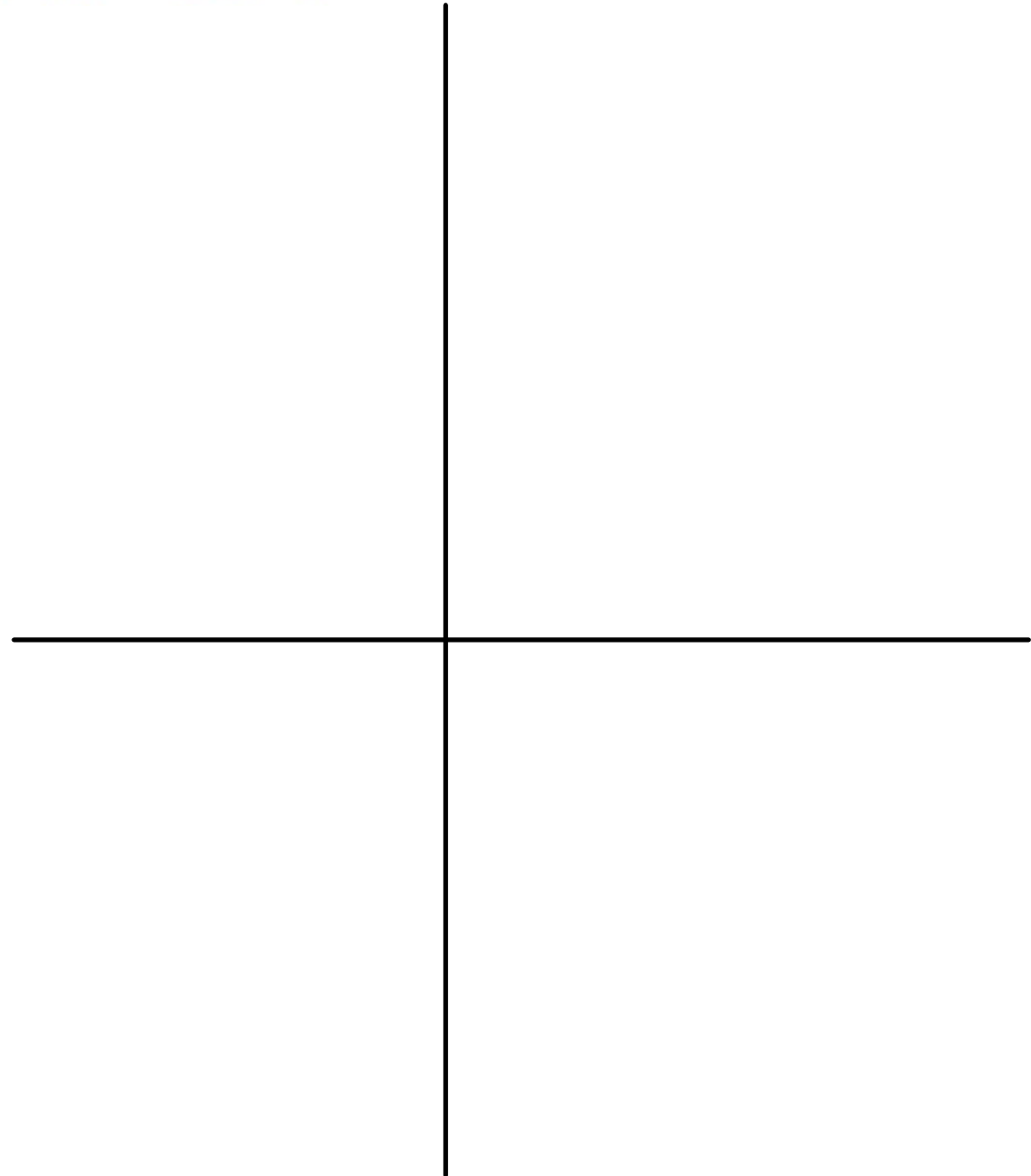


Evaluate the following integrals over the region stated, by changing to polar coordinates.

1. $\iint y^2 dx dy$ over the area outside $x^2 + y^2 - ax = 0$ and inside $x^2 + y^2 - 2ax = 0$

$$x^2 - ax + \frac{a^2}{4} + y^2 = 0$$

x



2. $\iint \frac{(x^2+y^2)^2}{x^2y^2} dx dy$ over the area common to $x^2 + y^2 = ax$ and $x^2 + y^2 = by, a, b > 0$

$$x^2 - ax + y^2 = 0$$

$$x^2 - ax + \frac{a^2}{4} + y^2 = \frac{a^2}{4}$$

$$(x - a/2)^2 + y^2 = (\frac{a}{2})^2$$

$$(a/2, 0) = a/2$$

$$\theta: 0 \rightarrow \pi/2$$

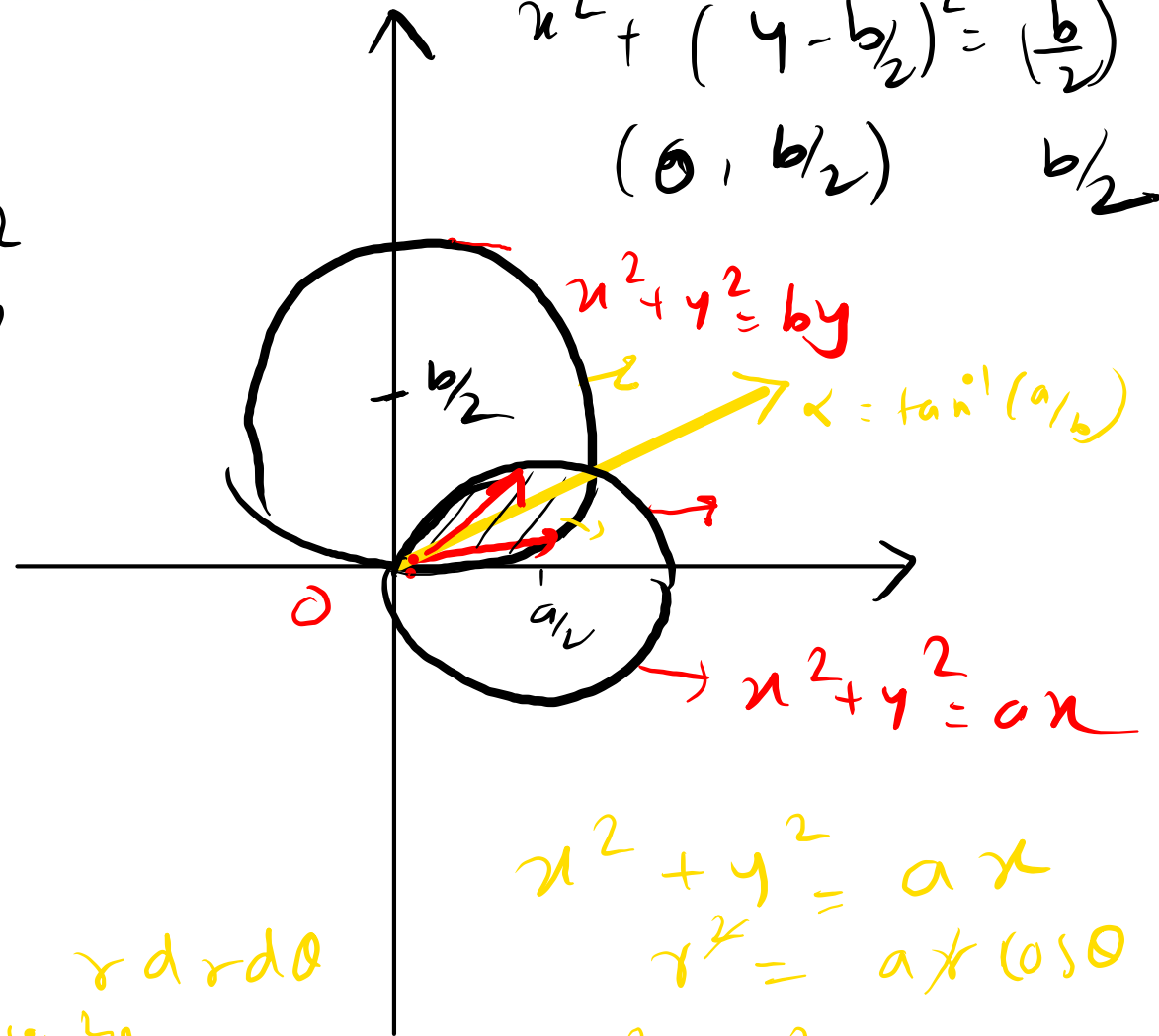
$$\int_0^{\pi/2} \int_0^{r=b \sin \theta} \frac{(r^2)^2}{r^4 \cos^2 \theta \sin^2 \theta} r dr d\theta$$

$$\int_{\pi/2}^0 \int_0^{r=a \cos \theta} \frac{r^4}{r^4 \cos^2 \theta \sin^2 \theta} r dr d\theta$$

$$\rightarrow x^2 + y^2 - by + \frac{b^2}{4} = \frac{b^2}{4}$$

$$x^2 + (y - b/2)^2 = (\frac{b}{2})^2$$

$(0, b/2)$ $b/2$



$$x^2 + y^2 = ax$$

$$r^2 = a r \cos \theta$$

$$x^2 + y^2 = by$$

$$r^2 = b r \sin \theta$$

$$a \cos \theta = b \sin \theta$$

$$\tan \theta = a/b$$

$$\theta = \tan^{-1}(a/b) = \alpha$$

3. $\iint_R \sqrt{a^2 - x^2 - y^2} dx dy$ where R is the area of the upper half of the circle $x^2 + y^2 = ax$

$$\int_0^{\pi/2} \int_0^{a \cos \theta} \sqrt{a^2 - r^2} r dr d\theta$$

$$x^2 + y^2 = ax$$

$$x^2 - ax + \frac{a^2}{4} + y^2 = \frac{a^2}{4}$$

$$(x - a/2)^2 + y^2 = (a/2)^2$$

center $(a/2, 0)$
 & rad $a/2$

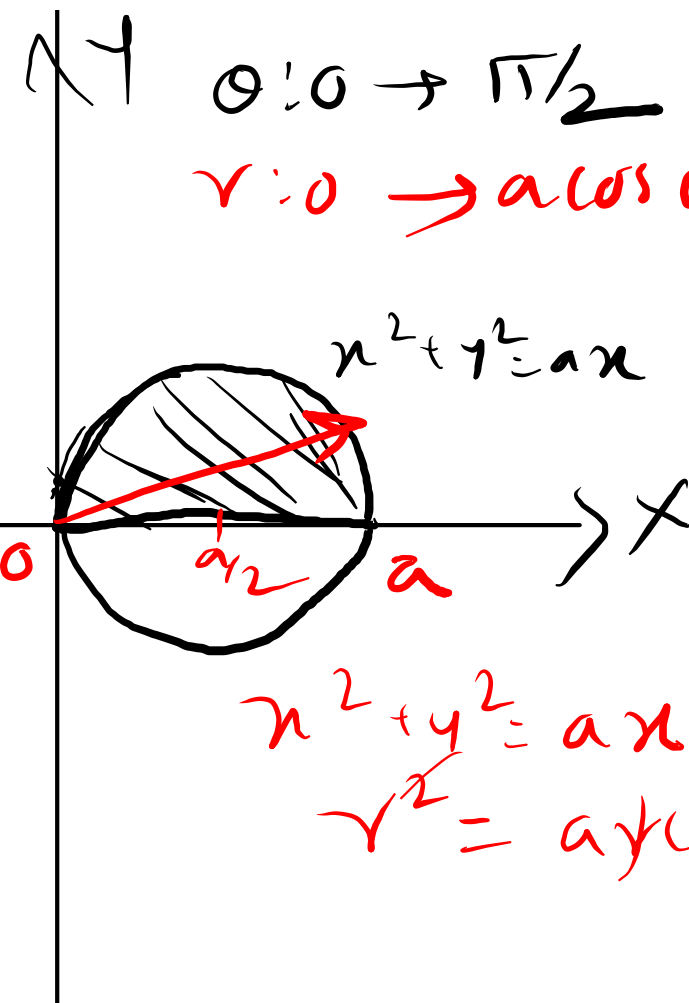
$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$dx dy = r dr d\theta$$

$$\theta: 0 \rightarrow \pi/2$$

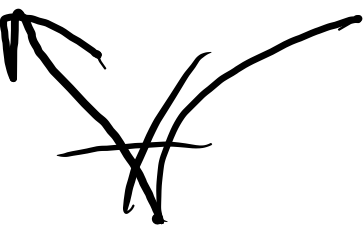
$$r: 0 \rightarrow a \cos \theta$$



$$x^2 + y^2 = ax$$

$$r^2 = ax \cos \theta$$

$$\frac{\pi/2}{4f}$$



2

∞

7. $\iint \frac{1}{(1+x^2+y^2)^2} dx dy$ over one loop of the lemniscate $(x^2+y^2)^2 = x^2 - y^2$

polar

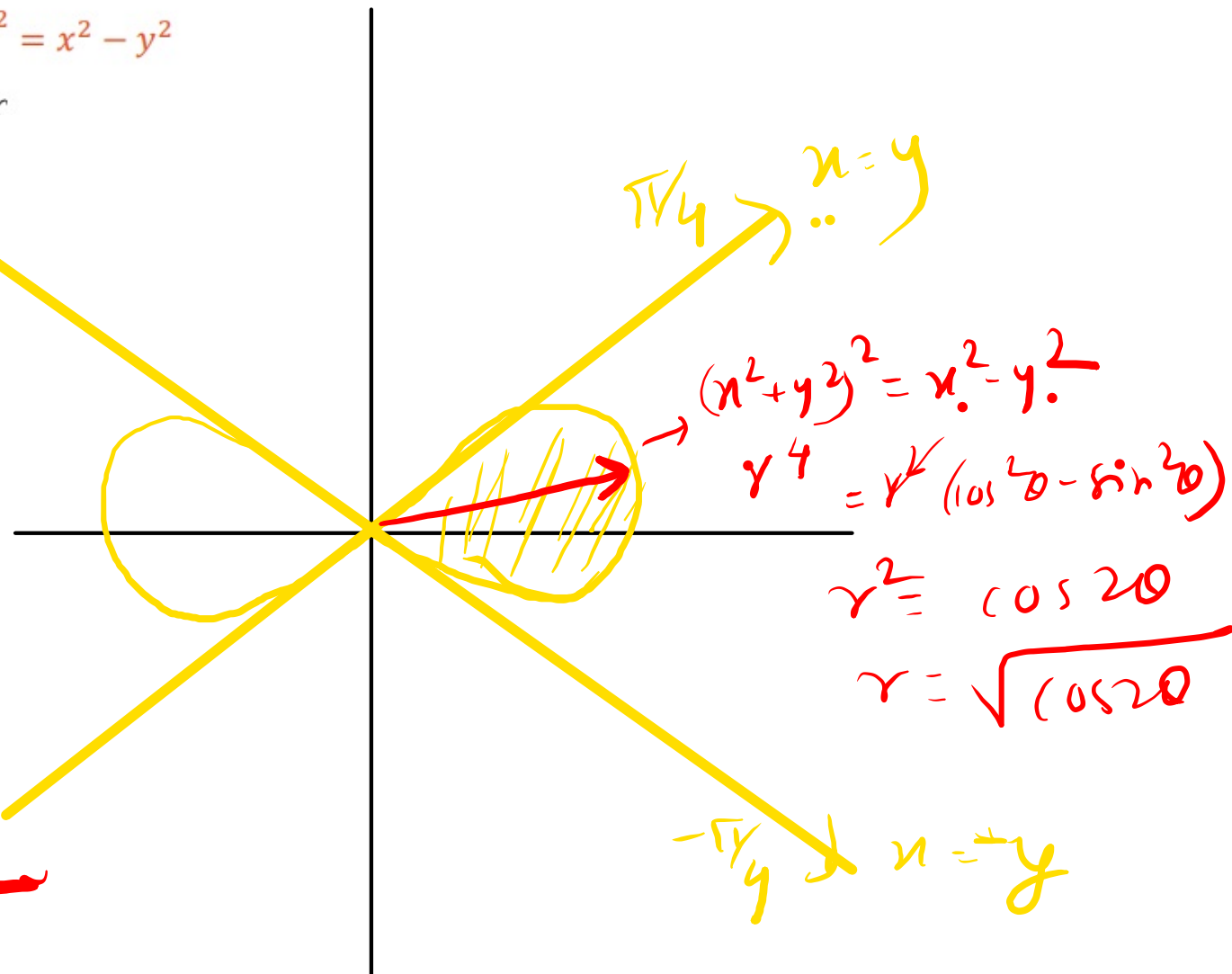
$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$dx dy = r dr d\theta$$

$$\theta: -\pi/4 \rightarrow \pi/4$$

$$r: 0 \rightarrow \sqrt{\cos 2\theta}$$



8. Evaluate $\iint_R (3x + 4y^2) \, dx \, dy$ where R is the region in the upper half of the area bounded by the circle $x^2 + y^2 = 1, x^2 + y^2 = 4$

